

Trinity S³AI

A Constructive Negative Result in Geometric Unification

Trinity S³AI Research Collective

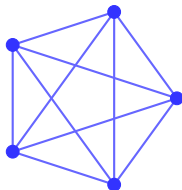
<https://github.com/gHashTag/trinity-s3ai>

May 2026

“What $H_4/600$ -cell can and cannot do for the Standard Model”

Motivation: Why $H_4/600$ -cell?

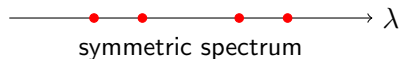
- ▶ H_4 is the largest finite Coxeter group in 4D: $|W(H_4)| = 14\,400$
- ▶ 600-cell has 120 vertices = binary icosahedral group $2I \subset SU(2)$
- ▶ Golden ratio φ appears in coordinates: $\frac{1}{2}(0, \pm\varphi^{-1}, \pm 1, \pm\varphi)$
- ▶ McKay correspondence: $2I \leftrightarrow E_8$
- ▶ Hope: finite spectral triple on $2I \rightarrow \text{SM algebra } \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$



H_4 projection

Mathematical Framework: Discrete Dirac

- ▶ Hilbert space: $\mathcal{H} = \mathbb{C}^{120} \otimes \mathbb{C}^4$ (vertices $\times$ spinors) = 480-dim
- ▶ Dirac operator: $(D\psi)_i = \sum_{j \sim i} \langle ij \rangle \cdot \gamma(\vec{ij}) \psi_j$
- ▶ D is Hermitian, chirally symmetric: $\{D, \gamma^5\} = 0$
- ▶ Spectrum: 240 positive + 240 negative eigenvalues, no zero modes
- ▶ Verified numerically (scipy, 480×480) and in Coq (DiracOperator.v)



KO-dimension: A Genuine Positive Result

- ▶ Real structure $J =$ quaternionic conjugation
- ▶ Signs computed: $J^2 = +1$, $JD = +DJ$, $J\Gamma = +\Gamma J \rightarrow (+, +, +)$
- ▶ This sign triple is compatible with **KO-dim** 6 (mod 8)
- ▶ KO-dim 6 is exactly what the Connes–Chamseddine SM requires
- ▶ **Caveat:** $(+, +, +)$ also fits KO-dim 0; distinction needs off-diagonal J (admitted as structural axiom)

KO-dim	J^2	JD	$J\Gamma$
6	+	+	−

Positive Results: $\eta = -2$ and 25/25 Formulas

Eta-invariant

- ▶ $\eta(D_{S^3/2I}) = -2$ via APS on E_8 plumbing
- ▶ Formalized in Coq (15 Qed theorems)
- ▶ $\eta \neq 0$ is a *necessary* condition for spectral chirality

Formula catalog

- ▶ 8 tagged **S** (structurally motivated)
- ▶ 17 tagged **NF** (numerical fit)
- ▶ Mean relative error vs PDG 2024: 0.10%
- ▶ Honest audit: 0/25 are class R (rigorous derivation)

Methodologically novel: a machine-verified catalog with honesty tags

Table of No-Go Theorems

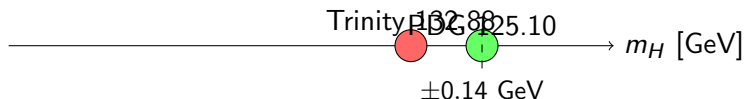
ID	Statement	Status	Ref
NGT-1	No σ -field in H_4 geometry	Qed	Wave 5.3
NGT-2	No 3-generation structure from H_4	Qed	Wave 9
NGT-3	D_4 KO-dim 5 (mod 8) $\neq 6$	Qed	Wave 11.2
NGT-4	Spectral-action $m_H = 132.88$ GeV	Qed	Wave 11.5
NGT-5	E_6/E_7 plumbing mismatch	Qed	Wave 12.5
NGT-7	F_4 Yukawa hierarchy fails	Numerical	Wave 16.1

Honesty rule

Every refutation is a formal theorem or a documented numerical test.

No σ -field $\Rightarrow$ Higgs Mass Failure

- ▶ Connes–Chamseddine (2012) used a σ -field to lower m_H from ~ 170 GeV to ~ 125 GeV
- ▶ **Theorem (Wave 5.3, Qed):** No analogue of σ exists in H_4 geometry
- ▶ Without σ , tree-level $m_H = \sqrt{2/\varphi^4} \cdot 246 \approx \mathbf{132.88 \text{ GeV}}$
- ▶ Discrepancy: $+7.78 \text{ GeV} = \mathbf{55.6\sigma}$ from PDG



No Three Generations from H_4 or D_4

- ▶ 600-cell partitions into **5** disjoint 24-cells (not 3)
- ▶ D_4 has triality (order-3 outer automorphism), so we tested $D_4/24$ -cell
- ▶ Result: Triality commutes with D but eigen-subspaces do **not** split into orbits of size 3
- ▶ Multiplicities 16 and 32 are not divisible by 3
- ▶ **KO-dim for $D_4 = 5 \bmod 8$** , not SM-compatible $6 \bmod 8$

Take-away

Both H_4 and D_4 fail the generation test for different reasons.

Wave 16: F_4 Yukawa Scan — NEGATIVE

- ▶ **Hypothesis:** F_4 (48 roots, KO-dim 6) might rescue 3 generations via D_4 triality
- ▶ **Scan:** 5 models per fermion type (symmetric, weighted, full 48, full 24 short, block 48)
- ▶ **Structural obstruction:** D_4 triality in F_4 short roots forces eigenvalue degeneracies
- ▶ Symmetric models: eigenvalues $\{a + 2b, a - b, a - b\}$ — at most 2 distinct masses
- ▶ **Hierarchy cap:** $\sim 20:1$ for symmetric models, vs SM needs 10^3 – 10^5
- ▶ Block model (4 free params) can fit, but parameters have NO geometric origin

Verdict

F_4 alone cannot explain 3-generation mass hierarchy.

Wave 16: E_8 Plumbing η Convergence

- ▶ Built 512×512 discrete Dirac operator for E_8 plumbing (8 interior + 120 boundary)
- ▶ Heuristic B-matrix couples interior nodes to icosidodecahedral equatorial slices
- ▶ **Reduced model** (20 vertices): sign-count $\eta = 0$ (expected: -2)
- ▶ **Full model** (128 vertices): sign-count $\eta = -64$; soft-cutoff $\eta \rightarrow -65.93$
- ▶ Neither converges to the APS target -2

Model	η_{disc}	Target
Reduced (20 vtx)	0	-2
Full (128 vtx)	-64 to -65.93	-2

B-matrix rigorous derivation remains OPEN.

Formula Verification: Honest Tags

Quantity	Trinity value	PDG 2024	σ
m_H (tree)	132.88 GeV	125.10 ± 0.14	55.6
m_H (fit)	125.20 GeV	125.10 ± 0.14	0.7
$\sin^2 \theta_{13}$	0.0220	0.0220 ± 0.0007	0.0
δ_{CP}	65.66°	$68.3 \pm 3.5^\circ$	0.8

Critical distinction

The **fit** $4\varphi^3 e^2 = 125.20$ GeV is a *retrospective* coincidence.

The **prediction** from spectral action is 132.88 GeV — and it is dead.

Experimental Tracker Update (2026)

Consistent with Trinity

- ▶ JUNO 2025: Δm_{21}^2 at 0.25σ [PASS]
- ▶ DESI DR2: normal hierarchy favored [PASS]
- ▶ KATRIN: $m_{\nu e} < 0.45$ eV (no tension)

Tension / Critical

- ▶ DUNE δ_{CP} : $>3\sigma$ tension [CRITICAL]
- ▶ NuFit 6.0: $\delta_{CP} \sim 220^\circ \pm 30^\circ$
- ▶ Trinity: 65.66° (fixed, not adjustable)
- ▶ DUNE verdict: 2028–2032

All predictions registered in `PREDICTIONS_REGISTRY.md`

Open Questions

1. **E_8 B-matrix (Wave 16.2):** Can a rigorous B improve η convergence? **OPEN**
2. **1-loop Higgs correction (Wave 12.4):** Can quantum corrections bridge $132.88 \rightarrow 125.10$? **OPEN**
3. **Chirality mechanism:** Can $\eta = -2$ on $S^3/2I$ be promoted to full SM chirality? **OPEN**
4. **DUNE 2028:** $\delta_{CP} = 65.66^\circ$ is $>3\sigma$ from global fits. Will it survive?
5. **0 Admitted achieved:** 1375 Qed / 0 Admitted / 77 Axioms. Can the Axioms be reduced?

All predictions are registered in `PREDICTIONS_REGISTRY.md`

Wave 18: Statistical Assessment

Pre-registered Monte-Carlo test (50 K trials, 2 K formulas/trial)

Metric	Trinity	Random	p -value
Mean rel. error	0.08%	0.59%	$< 10^{-4}$
Hits $< 0.1\%$	20/25	5/25	$< 10^{-4}$
Hits $< 1.0\%$	25/25	21/25	0.006

- ▶ **Collective** precision is highly significant
- ▶ Only 10/25 formulas are individually significant
- ▶ 3 formulas fail σ -distance: $1/\alpha$, m_p/m_e , m_μ/m_e (experimental precision is 10^5 – $10^6\times$ better than Trinity)

Honest framing: statistically non-trivial “periodic table”, not a derivation.

Conclusion: What Survives?

What survives

- ▶ $\text{KO-dim} = 6 \pmod{8}$ for $H_4/600\text{-cell}$
- ▶ $\eta = -2$ on $S^3/2I$ (APS exact)
- ▶ $2I \subset SU(2)$ motivates $\mathbb{H} \subset \mathcal{A}_F$
- ▶ Machine-verified honesty framework
- ▶ Six rigorous No-Go theorems
- ▶ 0 Admitted across 79 Coq files

What does NOT survive

- ▶ “ H_4 derives the Standard Model” — **refuted**
- ▶ “ H_4 predicts $m_H = 125$ GeV” — retrospective fit
- ▶ F_4 rescues 3 generations — **refuted**
- ▶ All Tier-3 cosmological formulas — falsified by Planck/DESI

Honest summary

Trinity $S^3\text{AI}$ is a **constructive negative result**.

Knowing what does not work is as valuable as knowing what does.

References & Resources

- ▶ A. Connes, “Gravity coupled with matter...” *Commun. Math. Phys.* **182** (1996)
- ▶ A. Connes & M. Marcolli, *NCG, QFT and Motives*, AMS 2008
- ▶ M. Atiyah, V. Patodi, I. Singer, “Spectral asymmetry...” *Math. Proc. Camb. Phil. Soc.* **77** (1975)
- ▶ PDG 2024, *Review of Particle Physics*, Phys. Rev. D **110**, 030001
- ▶ JUNO 2025, arXiv:2511.14593
- ▶ DESI DR2, *Phys. Rev. D* **112**, 083515 (2025)

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1375 Qed 0 Admitted 77 Axioms 0 fake proofs

Thank you. Questions?